

Generative Agents in Agent-Based Modeling: Overview, Validation, and Emerging Challenges

Carlo Adornetto , Adrian Mora , Kai Hu , Leticia Izquierdo Garcia , Parfait Atchade-Adelomou ,
Gianluigi Greco , *Member, IEEE*, Luis Alberto Alonso Pastor , and Kent Larson 

Abstract—The advent of generative agents (GAs) based on large language models (LLMs) has significantly influenced the evolution of agent-based modeling (ABM), offering new perspectives across various domains, including engineering and social sciences. This article provides an extensive overview of the integration of GAs into ABMs, emphasizing the advancements and emerging challenges in their validation. Traditional ABMs, characterized by their simplistic yet powerful approach to modeling complex systems, have been redefined with the introduction of GAs. This new generation of agents is often equipped with conversational capabilities. These agents, capable of simulating believable human behaviors and interactions, present unique opportunities and hurdles, especially in urban simulations and social dynamics. We explore the nuanced differences between traditional ABMs and ABMs populated by GAs—called GABMs. We delve into the state-of-the-art implementations of GAs, and review various validation methods. Through this comprehensive examination, we aim to shed light on the potential and limitations of GAs, advocating for the design of hybrid ABM-GABM approaches and systematic validation.

Impact Statement—Generative agents (GAs), powered by large language models, can emulate realistic human behavior in

agent-based models (ABMs). While they promise significant advancements in understanding social behavior, several technological challenges must be addressed to make GAs feasible, reliable, and believable in simulations. This article reviews this emerging research trend, emphasizing the validation of ABMs based on GAs. We specifically explore the differences between traditional ABMs and those enhanced by GAs. We also discuss key state-of-the-art GA architectures and the methods for validating their behavior. By collecting the most widely used validation methods for traditional ABMs in the literature, we advocate for a holistic validation platform that integrates both traditional and GA-specific methods to assess ABM validity.

Index Terms—Agent-based models, generative AI, generative agents, generative agent-based models validation, large language models.

I. INTRODUCTION

THE advent of large language models (LLMs) suggests we rethink the entire agent-based model (ABM) processes, from the design of new enhanced agents able to perceive, think, memorize, and act as likely as possible to humans [1], [2], to the methodologies and approaches by which we validate their behaviors. An emergent trend in research is the design of ABMs populated by GAs, where agents evolve and make decisions using LLMs, leveraging natural language interactions. Unlike traditional manually defined agents, LLMs offer greater flexibility and realism by encapsulating principles of human behavior, implicitly inherited from the vast amount of data they digest to learn. We can now refer to the definition provided in [3], by which GAs are “*Computational software agents that simulate believable human behavior.*”

This definition, while straightforward, lacks a formal interpretation of the agent’s believability, making it difficult to validate the interactions within the system—especially in systems that involve human-like conversational interactions. A commonly used approach to address this issue is to engage human validators who manually rank the agent’s behavior based on perceived believability [3], [4]. However, it is essential to note that the level of rigor needed for validation varies depending on the purpose of the ABMs. Such models are indeed approximations of the real world primarily designed to guide further research. In this context, qualitative validation of these systems, as opposed to quantitative ones, should be encouraged for the ABMs’ capacity to inspire additional empirical research and theoretical advancements.

By assuming that readers are already familiar with the ABM framework and the methods for their development and

Received 8 November 2024; revised 26 March 2025; accepted 25 April 2025. Date of publication 5 May 2025; date of current version 26 November 2025. This work was supported in part by the PNRR projects “FAIR (PE00000013) - Spoke 9” and “Tech4You (ECS00000009) - Spoke 6,” under the NRRP MUR program funded by the NextGenerationEU and in part by the project “STROKE 5.0 (B29J23000430005)” funded by MIMI Program num. F/310031/01-05/X56. This article was recommended for publication by Associate Editor Geoff Nitschke upon evaluation of the reviewers’ comments. (Corresponding author: Carlo Adornetto.)

Carlo Adornetto is with the Department of Mathematics and Computer Science, University of Calabria, 87036 Rende, Italy, and also with the City Science Group, MIT Media Lab, Massachusetts Institute of Technology, Cambridge, MA 02139-4307 USA (e-mail: carlo.adornetto@unical.it).

Adrian Mora is with the Universidad Politécnica de Madrid, 28040 Madrid, Spain (e-mail: a.mcarro@upm.es).

Kai Hu is with the South China University of Technology, Guangzhou 510641, China (e-mail: arhukai@scut.edu.cn).

Leticia Izquierdo Garcia, Luis Alberto Alonso Pastor, and Kent Larson are with the City Science Group, MIT Media Lab, Massachusetts Institute of Technology, Cambridge, MA 02139-4307 USA (e-mail: leticiat@media.mit.edu; alonsolp@media.mit.edu; kll@media.mit.edu).

Parfait Atchade-Adelomou is with the City Science Group, MIT Media Lab, Massachusetts Institute of Technology, Cambridge, MA 02139-4307 USA, and also with the Smart Society Research Group, La Salle, 08022 Barcelona, Spain (e-mail: parfait@media.mit.edu).

Gianluigi Greco is with the Department of Mathematics and Computer Science, University of Calabria, 87036 Rende, Italy (e-mail: break:sgianluigi.greco@unical.it).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TAI.2025.3566362>, provided by the authors.

Digital Object Identifier 10.1109/TAI.2025.3566362

validation, this work aims to provide an overview of the evolution of ABMs, which serve as computational models for simulating the actions and interactions of autonomous agents. In particular, we explore how the integration of GAs, enhanced by LLMs, introduces new dimensions to these models by enabling the simulation of more realistic and adaptive human behaviors.

We want to further emphasize that GABMs should not be seen as a mere innovative version of ABMs. Instead, they represent an opportunity to retain the advantages of ABMs while addressing some of their limitations. The most fundamental and significant limitation pertains to the reproducibility of human behavior. Traditional ABMs rely on mathematical frameworks or predictive models and typically assume predefined behavioral rules for agents' decision-making, often based on the assumption that individuals act rationally. However, behavioral science suggests that human behavior frequently deviates from pure rationality, exhibiting biases and bounded rationality in decision-making (e.g., in economics [5]). GABMs, involving GAs and, hence, driven by LLMs, offer a way to relax this strict rationality assumption. LLMs have the capability to replicate diverse and realistic behavioral patterns, allowing GABMs to simulate more believable decision-making processes that better reflect human complexity and variability, as shown for example in [3], [6], [7]. This characteristic makes GABMs particularly well-suited for capturing the often nonlinear, nonrational, and emergent behaviors exhibited by humans.

To support our thesis, we present an overview of the emerging GABM field, highlighting its potential and challenges. A key takeaway from this study is that both traditional ABM and GABM come with distinct strengths and weaknesses. Rather than positioning GABM as a replacement for ABM, this work offers a comparative analysis, identifying the advantages and limitations of both approaches. We emphasize that significant research is still needed to fully understand how to combine the best of both worlds, leveraging GABM's cognitive and adaptive capabilities while retaining the efficiency and interpretability of traditional ABM.

To review the state-of-the-art, we started by searching relevant keywords and their combinations on Google Scholar and DBLP, including "Generative Agent," "ABMs Validation," "Agent-based modeling," and "Generative ABMs." For the validation of traditional ABMs, we concentrated on highly cited journal reviews. However, while there was a lack of results specifically addressing the validation of ABMs that use GAs, we started with the most influential implementations of GAs and traced the reference chain.

This work is organized as follows: Section II highlights the differences between traditional ABM and generative ABM, providing definitions and terminology. Section III discusses state-of-the-art techniques used for designing agents able to evolve during simulations. In Section IV, we review state-of-the-art realizations of GAs underlying distinctive aspects of their architectures, summarized in Table II. Section V reviews state-of-the-art methods for validating ABMs, while Section VI explores the transition from ABM to GABM validation, highlighting prominent approaches in this emerging field and detailing existing validation techniques for GABMs. Section VII

examines the key advantages and limitations of both ABM and GABM, presenting a structured comparison in Table III that underscores their complementary nature. The final two sections, VIII and IX, discuss the challenges and the contributions of GABM to the traditional ABM research field, advocating for combined ABM-GABM approaches to address their limitations.

II. ABMS AND GENERATIVE ABMS

The word "*generative*" in the field of ABMs was introduced by Epstein in [8], and later extended in [9] referring to the ability of ABMs, framed into the field of social sciences, to generate emergent social behaviors. Epstein's work emphasized how ABM can be considered a generative approach to social science. He advocates for building models where social phenomena are "grown" from the interactions of individual agents. Indeed, his work is highly regarded for demonstrating how simple local interaction rules can generate complex social behaviors, making it a foundational text for those interested in the principles and practices of ABM. His interpretation of the field has inspired numerous subsequent works, where authors refer to generative ABMs as systems capable of producing behaviors from simple rules [10], [11], [12]. One illustrative example of generative ABMs is Schelling's segregation model [13], demonstrating how simple individual preferences regarding neighbors' similarity can lead to large-scale segregation patterns. In this model, agents with a slight preference for neighbors of the same type rearrange themselves over time, resulting in highly segregated spatial distributions, even when their preferences are not extreme. This emergent behavior exemplifies the generative power of ABMs, showing how microlevel decision rules can lead to unintended macrolevel outcomes. Another well-known generative ABM is Reynolds' Boids model [14], which simulates the flocking behavior of birds based on three simple rules: separation (avoiding crowding), alignment (matching velocity with neighbors) and cohesion (moving toward the center of nearby agents). Despite the simplicity of these rules, the model generates lifelike flocking dynamics without a central controller, demonstrating how local interactions can give rise to complex emergent behaviors. The Boids model has influenced various fields, including computer graphics, robotics, and behavioral ecology, making it one of the most famous generative ABMs.

These systems, although generative, do not necessarily involve the use of generative algorithms or generative Artificial Intelligence (AI) in their implementation. Hence, the contemporary use of the term "*generative*" can lead to ambiguity in this research field, particularly with the advent of generative AI, which can be used to develop enhanced agents or scenarios. In recent works, authors developed ABMs by relying on GAs, referred to as GABM [15].

The following aims to clarify the distinction between generative ABM and GABM to establish a systematic terminology.

A. Definitions and Purposes

Since its inception, ABMs have been used to simulate systems at a high level, capturing emergent properties and social

TABLE I
COMPARISON BETWEEN THE GENERATIVE AND NON-GENERATIVE VERSION OF THE SAME SIMULATION MODEL

Traffic Flow Simulations		
Consider an ABM designed to simulate traffic flow in a specific city. Agents in the model represent cars, and the model simulates how different traffic management strategies (e.g., traffic lights, road closures) affect congestion.		
	ABM	generative ABM
Objective	It may be useful for practical purposes, such as predicting congestion or visually analyzing traffic flow, but do not advance social science as a discipline. It is used merely as a computational tool to simulate specific scenarios or when the model's findings cannot be generalized or do not contribute to broader social theories.	Instead of merely optimizing traffic flow, this model aims to understand how social norms, individual decision-making, and collective behavior influence traffic patterns and urban mobility. The model seeks to uncover generalizable insights about the interplay between individual choices and societal outcomes.
Outcomes	While the model may produce valuable results for urban planners (e.g., identifying the best traffic management strategy), it is not necessarily generative of social science. The model is tailored to a specific context and does not contribute to developing new social theories or explaining human behavior. Its findings are specific to the simulated scenario and may not apply to other contexts.	Theoretical Insights: The model generates insights into how social norms and individual behaviors interact with urban infrastructure to produce complex traffic patterns. It helps explain how seemingly minor changes in individual decision-making can lead to significant shifts in urban mobility and social outcomes. Generalization: The findings from this model can be generalized to other contexts, providing insights into how social behaviors shape urban systems, how collective action problems (like traffic congestion) can be mitigated through social influence, and how policies can leverage social dynamics to achieve desired outcomes.

phenomena through simplistic models. This abstraction allows broad applicability. Authors in [16] propose a classification of ABMs into seven distinct modeling purposes: *Explanation, Prediction, Description, Theoretical Exposition, Illustration, Analogy, Social Learning*. Using these a basis for our discussion, we assert that while all of these ABMs can be implemented by using GAs, the generative social science nature of the models depends on their purposes and the way scientists explain their results [16]. The presence of GAs is not a sufficient condition to make a system generative. According to the definitions provided by [17], models are considered generative of an observed social phenomenon when they describe it in terms of the external (environmental and social) and internal (behavioral) mechanisms that generate it, rather than inferring causes from observed covariations—running the systems under different conditions and studying the variations. Hence, only the ability to explain behavior in terms of emerging internal processes makes an ABM system generative. In Table I, we propose an example in the context of city simulation, in which a generative version of a traffic flow ABM is compared to a nongenerative one.

Under this definition, models designed for *Explanation, Theoretical exposition* or *Social Learning* purposes are for their nature generative of social science. On the other hand, models designed for *Prediction, Description, Illustration*, or *Analogy* are not necessarily generative. Therefore, we argue that the focus, when referring to generative ABM, should be on the generative properties of a system rather than the modeling purposes or the mere presence of GAs. Additional details on model purposes and examples are provided in the Supplementary Material.

Aligning with current literature, in the following, we will use the term ABM for the broader class, encompassing generative (of social science) ABMs and the models designed with rule-based, evolutionary, or reinforcement learning approaches. For ABMs specifically designed with GAs, we will use the acronym GABMs.

III. EVOLVING AGENTS

Research in ABM, over the last decades, has focused on how to enhance agents and systems capable of adaptation and evolution, which aligns with GABMs' objectives.

For this purpose researches have focused on bio-inspired evolutionary algorithms (EA), and machine learning (ML)—particularly reinforcement learning (RL) in ABMs—enhancing agents' evolution in beyond simple rule-based behaviors. For a detailed review and examples of bio-inspired and RL applications, see Supplementary Material.

AEs, inspired by the Darwin's theory of evolution, mimic natural selection by refining agent behaviors through selection, mutation, and recombination [18], [19]. EAs encode solutions as individuals, evolving them via fitness-based selection [20]. *Genetic algorithms* use crossover and mutation to optimize agent strategies in simulations [21], while *evolution strategies* focus on self-adaptive mutation for continuous parameter optimization in dynamic ABMs [22], [23]. Other particularly interesting evolutionary approaches are based on *cultural algorithms* [24], integrated in ABMs for urban planning and artificial societies [25], [26]. They extend the evolutionary paradigm by incorporating a shared *belief space*—manually defined by developers—representing a higher-level knowledge repository, which stores knowledge extracted from the population's experiences to guide the evolution, adaptation and behavior of individual agents [27].

Despite being widely used, AE-based ABMs face several limitations. One key challenge is computational complexity—EAs often require significant computational resources due to their iterative nature. Moreover, EAs can prematurely converge to local optima, limiting their ability to discover diverse or truly optimal strategies, especially in dynamic environments requiring adaptation. In addition, their performance depends heavily on fine-tuned hyperparameters like mutation rates and selection pressure.

In the specific case of cultural algorithms, continuous interactions between the *belief space* and agents increase computational demands. Moreover, designing an effective belief space is challenging—poor choices can mislead agents, cause premature convergence, and reduce behavioral diversity, weakening system robustness. Additionally, extracting knowledge from agents' experiences is complex and prone to biases. For example, in traffic simulation, premature convergence might cause all agents to avoid a specific street, inadvertently creating new

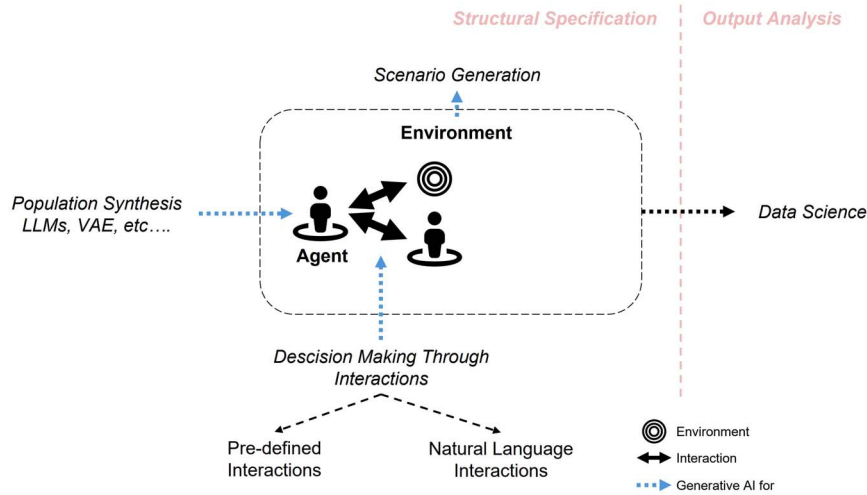


Fig. 1. AI can be used in two distinct areas of ABM development: Structural specification and output analysis. Over them, generative AI can contribute to population synthesis, decision making through interactions, and scenario generation.

bottlenecks elsewhere. As shown in [28], well-known shared routing strategies can deteriorate collective performances.

RL, on the other hand, is a subset of ML approaches that enables agents to learn by interacting with dynamic environments, refining behavior through trial and error while maximizing cumulative rewards [29], [30]. In the field of ABM, RL has been used in a wide spectrum of simulations in the contexts of pandemic emergencies [31], traffic management [32], [33], residential land growth [34], and economic systems [35].

An example of RL in human-like behavior learning is pedestrian movement simulation. Agents navigate spaces like train stations, optimizing mobility by observing surroundings and adjusting speed or direction. Rewards incentivize efficient travel while penalizing collisions and delays. Initially moving randomly, agents learn over time to avoid congestion and choose optimal paths, improving navigation efficiency.

According to [36], the popularity of RL in behavioral modeling for ABMs can be attributed to factors such as the realistic modeling of human decision-making processes and the agents' ability to learn from both broader system dynamics and agent-agent interactions. On the other hand, RL intrinsic discrete nature presents several limitations. RL relies on predefined features, rewards, and rules to shape agent behavior, which demands for developers' expertise and domain specialists. This restricts flexibility and often leads to oversimplified representations of complex decision-making processes. For instance, in scenarios such as pedestrian movement during a pandemic (mentioned above), agents maintaining distance based on fear must be encoded with predefined reward functions. Such rigid formulation limits generalization and constrain adaptability beyond explicitly programmed responses, making agents inherently task-specific. While this allows for optimization within well-defined tasks, it lacks the dynamic reasoning and adaptability needed to emulate naturalistic behaviors such as negotiation, deception, persuasion, and cultural adaptation.

The next sections examine how GAs, assuming their ability to generate believable human behavior [3], [37], [38], can overcome limitations of EA-based and RL-based ABMs. While scalability and computational demands remain challenges, GAs offer a flexible way to evolve memories and beliefs without predefined retention rules or hyperparameters tuning, enabling agents' contextual decision-making.

However, our aim is not to claim superiority of one approach over another but to highlight their complementary strengths. For instance, combining RL with GAs in ABMs has shown great promise. Early experiments integrating RL algorithms with language models for interaction simulation have already produced encouraging results [39].

A. Generative AI in ABM

Two primary areas of ML application can be identified in ABM: *structural specification* and *output analysis*. In this work, we will focus on *structural specification* concerning the integration of Generative AI to model agents' decision-making and interactions. In the following, we discuss this integration providing a foundational understanding of what we mean when referring to GAs. Fig. 1 visualizes how generative AI can be used in three key areas

- 1) *Population synthesis*: Generative AI can create synthetic datasets of agent characteristics and plans (e.g., in [40]). In this case, even though a generative AI model generates agents and their behaviors, it is not accurate to refer to them as GAs, as the evolution of agents within the system does not leverage generative processes, and their behavior is defined ex-ante.
- 2) *Decision making through interactions*: Generative AI can model interactions between agents and between agents and their environment. These interactions can be modeled through simple inference or systems that allow agents to interact using natural language (e.g., [3]). In the latter case, a generative algorithm generates decisions

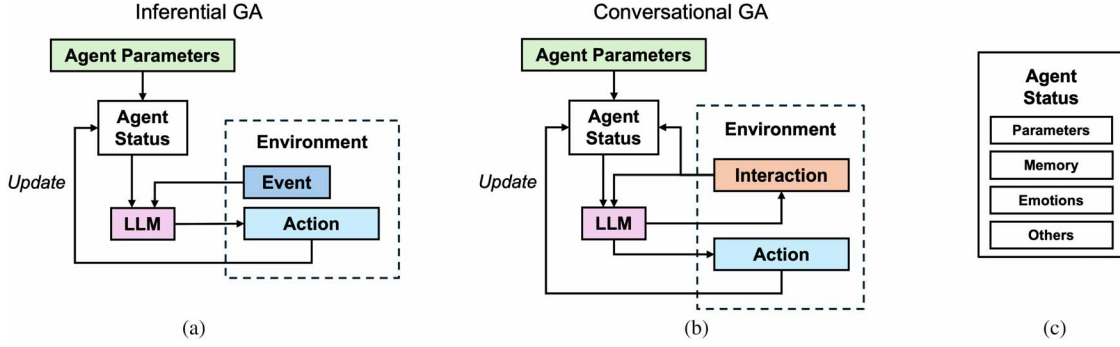


Fig. 2. LLM-based feedback loop and status of GAs. (a) Inferential GA feedback loop where the agent’s LLM receives the status of the agent and an event as inputs; LLM produces an action; such action updates the agent’s current status. (b) Conversational GA feedback loop where the LLM decides based on the agent’s status and natural language interactions. Such interactions can directly affect the status. (c) Example of an agent status content.

dynamically, considering the evolving history, making it appropriate to refer to these agents as GAs.

- 3) *Scenario generation*: Generative AI can aid in creating and characterizing environments (e.g. [15]).

In *Decision Making Through Interactions*, the advent of LLMs as generative AI models have inspired new simulation paradigms, based on the design of artificial agents that can exhibit human-like behavior [2]. We refer to such agents as GAs [3], [41].

LLMs, with near-human text comprehension and generation, enhance ABM agents by simulating human decision-making, planning, and interactions.

IV. GENERATIVE AGENTS

As previously mentioned, we refer to GAs as a type of ABM agent that, powered by an LLM, can exhibit human-like behavior and interact with other agents, reacting to evolving scenarios.

To enable such capabilities, researchers can design GAs based on the concept of *feedback loop*, which improves individual decision-making using environmental data and historical information from evolving simulations and interactions. As proposed by [41], this feedback loop involves the environment and agents exchanging feedback about actions and current states with LLMs. In more detail, a mechanical model—a rule-based environment defined by developers—captures the system’s state, provides information to the LLM, and uses LLM-informed decisions to shape the system’s state. A notable feature of this feedback loop is that decision rules emerge from the extensive knowledge embedded in the LLMs rather than being imposed by modelers.

Based on the contribution from [41], we propose an extension of the feedback loop by focusing on GA’s decision-making processes. In particular, as shown in Fig. 2, we distinguish between the *inferential* case, in which LLMs are only used as decision-makers [Fig. 2(a)], and the case in which GAs are equipped with *conversational* capabilities, hence when LLMs are used to enable natural language interactions between GAs [Fig. 2(b)].

When GAs are developed with an inferential approach, agent decision-making is confined to a discrete GA characterized by

agent parameters. It holds agent status in both case regimes, where LLMs generate actions based on ongoing conversation dynamics and environment perception.

A. Parameters and Status

Agent parameters characterize a GA and holds agent status in both cases. Agent parameters typically include demographic information that identifies personas, such as high-level data like age, sex, income level, race, employment, and location, as well as social information detailing the relationship between agents [42]. This information, when combined with the extensive knowledge embedded in LLMs, can guide the decision-making process of an agent to emulate a human-like personality accurately. Agent status, on the other hand, represents the current state of the agent within the evolving simulation [see Fig. 2(c)], and in particular, it can be seen as the GA representation itself, excluding its decision-making processes (which are implemented by the LLMs). The status can be characterized differently depending on the application, potentially involving parameters and an evolving memory updated during the simulation. It may even include representations of a discrete emotional state. These sentiment states can be generated by the LLMs themselves, as demonstrated in the work by [43], which employs natural language processing techniques to perform frequent global measurements of emotional states through sentiment, assessing the impacts of the pandemic.

LLMs can exploit the information contained in the agent status to enhance and magnify the GA’s decision-making processes, which can vary according to different human-like personalities. A major practical advantage of using LLMs in this context is that all the information contained in the agent status during a simulation can remain in the form of structured text, which LLMs can directly interpret for decision-making in GAs. This approach is particularly relevant in peer-to-peer social influence and cultural diffusion (e.g., [44], [45]), a field traditionally explored using only classical ABMs. This domain provides a compelling use case for GAs, as it involves decentralized interactions where agents exchange cultural values, opinions, and behaviors without a central authority. In this

scenario, agents represent individuals engaging in social discussions, influencing each other's perspectives through natural language interactions. We show below a practical example of agent status representation:

Parameters: name: Micheal Russ; age: 32; nationality: USA; innate traits: friendly, charismatic, open-minded.
Memory: Micheal Russ is living with his wife, Kyla Russ, who is a college professor, and son, Eddy Russ, a philosophy student; Micheal Russ loves his family very much; Micheal Russ is uncertain about whom to vote for in the next election; Micheal Russ attended university with his friend Sam Smith, a strong Republican supporter; Micheal Russ had a conversation with Sam Smith yesterday about the upcoming election; Micheal Russ believes that Sam is trying to convince him to vote Republican.
Emotional status: CurrentEmotion: Skeptical; EmotionHistory: [Neutral, Curious, Skeptical]; RecentInteraction: "Discussed with Sam Smith about new elections, and got curious but skeptical for Sam Smith trying to convince him"; SentimentScore: 0.4.

In this example, the agent parameters include social attitudes and personality traits, which can influence how the agent interacts with others. The agent status captures the agent's evolving opinions, recent conversations, and emotions. Emotional states can be encoded in multiple ways, from discrete numerical scores to categorical representations or even as detailed natural language descriptions of experiences and resulting feelings.

As agents interact in natural language to discuss topics such as fashion trends, political views, or social practices, their perspectives evolve dynamically. If a sufficient number of agents align on a new behavior, this change propagates throughout the network, leading to the emergence of collective cultural patterns. Beyond the practical advantages of a straightforward natural language representation, this tries to shed lights on how LLM-powered GAs can enhance traditional ABMs by enabling more nuanced, human-like decision-making based on ongoing dialogue and memories.

B. Existing GAs

In this section we explore state-of-the-art applications of GAs for GABMs. A comparative summary of the following GABMs is provided in Table II.

Concerning the inferential GAs, authors of [47] proposed an interesting implementation. The authors developed an epidemic model that integrated human behavioral dynamics in response to evolving outbreaks by defining an inferential LLM-based architecture. These agents effectively mirrored human actions, exhibiting adaptive behaviors such as self-quarantining when unwell and isolating as cases escalated. A crucial contribution of this work is the systematic method used for defining personas, where agents' personalities are defined on an Agent Status based on the Big Five personality traits by [48] from the field of psychology. This study is a good example of how personality traits rooted in psychology literature could influence the LLM's responses.

More recent studies concern the development of conversational GAs to be used in GABMs.

An initial breakthrough in this field was recently achieved by Park et al. [3]. In this work, the authors proposed an original architecture for GAs to design entities that emulate believable human behavior in interactive settings. These GAs can carry

out everyday activities like eating breakfast, working, and conversing with each other. Such innovative agents' architecture strongly relies on natural language processing and generation in its key components: *memory streams*, *reflection*, and *planning mechanisms*.

The *memory stream* is a module that records experiences in natural language. To query such a memory, the authors designed a retrieval function based on the triple {*Recency*, *Importance*, *Relevance*}. This module enables the GA to reflect on past events and plan future actions based on their accumulated experiences. The *reflection* is a system for incorporating reflective knowledge synthesis within the architecture. This feature empowers the GA to generalize and draw higher-level inferences from their memories. Agents are equipped with a reflection module to generate trees (hierarchy) of reflections where the leaf nodes are base observation, and the nonleaf nodes represent thoughts that become more abstract and higher-level as they ascend the tree, summaries of the previous experiences. Lastly, the authors designed a *planning and reaction* mechanism based on the *reflection* and the current situation of surrounding GAs and the environment. These mechanisms enable the GA to create plans and responses that are consistent and appropriate over time and the autonomously evolving scenario.

A key characteristic of Park's GA architecture is that it couples LLM capabilities and a continuously updated memory stream that stores agents' experiences. Hence, in a continuous regime, items in the memory stream are retrieved to guide the LLM, generating actions, plans, and reflections.

Furthermore, the authors developed a virtual sandbox environment inspired by "The Sims" in which they simulated agents' behavior. In particular, 25 agents were observed and interacted with by testing for their individual and social behaviors (e.g., planning a Valentine's Day party autonomously). The simulation showed remarkable agent autonomy, coherence, and believability.

The work from Park et al. [3] has been a pioneering and disruptive contribution to the GABM and computational social sciences field and is highly inspiring for the research community.

By relying on Park's GA architecture, Vezhnevets et al. [15] recently introduced Concordia, a library that facilitates the development of language-mediated GABMs based on LLMs, to provide agents with more sophisticated cognitive capacities. These capacities enable the agents to perform actions grounded in physical, social, or digital spaces with complexity and realism that traditional ABMs cannot achieve. Concordia GAs consist of associative memories inspired by [3] and flexible components that mediate between LLM calls and memory retrieval. The agent status is equipped with what the authors call *components*, which are statements in natural language expressing the current state of an agent in terms of identity, plans, observation, and call for action. Each of these statements serves as a base knowledge to generate context for decision-making. All interactions between GAs are facilitated by a unique game master (GM) whose behavior can be grounded by physical, chemical, financial, etc., mathematical models of the real world. The GM simulates the environment, handling the world's state,

TABLE II
COMPARISON OF EXISTING GABMS, HIGHLIGHTING THEIR SCENARIOS, AGENT CHARACTERISTICS, AND VALIDATION METHODS

Ref	Year	Title	Scenario	Contribution Highlights	Validation
[4]	2024	<i>Sotopia: Interactive evaluation for social intelligence in language agents.</i>	Conflicts and cooperation goal-driven social interactions between GAs.	Open-ended simulation environment with GAs for different profiles and behavior dynamics.	Humans+LLMs holistic evaluation through human judgments vs LLM-based automatic validation.
[37]	2024	<i>Emergence of social norms in generative agent societies: principles and architecture.</i>	Social norms emergence within the environment from [3] using 10 GAs.	CRSEC architecture for: Creation & Representation, Spreading, Evaluation, Compliance for normative behavior establishment.	Humans - 30 evaluators, demonstrating effectiveness in reducing conflicts and establishing public and self norms for the GAs.
[15]	2023	<i>Generative agent-based modeling with actions grounded in physical, social, or digital space using concordia.</i>	Social interaction in grounded environments (e.g., small town economy, social experiments).	Introduction of GABMs general purpose framework for flexible, language-mediated interactions.	No Validation framework provided. Best practices and recommendations for the design of validation frameworks are discussed.
[3]	2023	<i>Generative agents: Interactive simulacra of human behavior.</i>	Social interaction in a small city between 25 agents, forming relationships, and planning activities.	New GA architecture with memory stream, planning, and reflection modules for behavior simulation.	Humans 100 evaluators for ablation studies and agent's behavior believability scoring.
[46]	2023	<i>S3: Social-network simulation system with large language model-empowered agents.</i>	Social interaction in online social network to study behaviors and propagation of information.	Fine-tuning and prompt engineering to ensure agent's behavior emulates humans on key aspects: emotion, attitude, and interaction.	Empirical validation through real-world social network data, metrics assessing accuracy at individual and population levels.
[47]	2023	<i>Epidemic modeling with generative agents.</i>	Epidemic scenario and decision-making in emergency dynamics to study human behavior.	Inferential GAs used for decision-making with no conversational abilities.	Visualization to demonstrate agents' behaviors aligns with real-world epidemic patterns.

agent actions, and interactions between GAs. It generates event statements representing agent actions' outcomes, mediates interactions, and updates environmental variables.

Another interesting work, again inspired by Park's agents [3], was recently proposed in [49]. They designed a new agent architecture called Lyfe, which relies on a sophisticated memory system and hierarchical action selection component. The latter allows GAs to be cost-effective by delegating only necessary decisions and interactions to LLM, avoiding the wide number of text generation calls usually needed to allow system continuity. As an even more interesting original contribution, and conversely to existing prominent GAs (e.g. [3], [50]), the Lyfe agents were provided the ability to entertain conversational interaction with multiple agents.

A recent extension of [3] has been also proposed by [37], framed in the context of the emergence of social norms in simulations. In particular, the authors propose the CRSEC architecture, which fosters the emergence of social norms within the sandbox environment from [3] by using 10 GAs Simulation. CRSEC operates through four key modules. The *Creation & Representation* module allows norm entrepreneurs to formulate personal norms and store them in their databases. The *Spreading* module facilitates norm diffusion, as some agents proactively influence others through conversations while others infer norms from interactions and observations. Identified norms undergo assessment in the *Evaluation* module, determining their viability. The *Compliance* module ensures that agents generate plans and actions aligned with norms, shaping social observations and reinforcing norm propagation. Over time, agents engage in memory synthesis to consolidate their personal norms.

To illustrate the effectiveness of the new architecture, the authors present a case study simulation in which an experienced smoker gradually internalizes the "no smoking indoors" norm

through interactions with other GAs and eventually enforces it by reminding them of the rule.

A comprehensive review of existing architectures for autonomous agents (not necessarily framed in the field of ABMs) has been recently provided by [51].

1) *Simulating Emotions: The Qualitative GABMs Dimension*: Another interesting implementation of GAs has been proposed by [46]. In this work, the authors designed an architecture for LLM-powered agents to simulate social network dynamics. These agents could observe content posted by other agents, adjust their attitudes and emotions, create their own content, or remain inactive. They successfully replicated the diffusion of information and the shifts in emotions and attitudes of users in response to two events: the release of nuclear wastewater in Japan and the incident involving the mother of eight children. Their approach shows promise for flexible behavioral modeling. However, in contrast with the GAs from [3], the agent's reasoning relies solely on the LLM without incorporating explicit computational mechanisms that represent human cognitive processes, such as memory retrieval based on relevance, recency, and frequency.

Interestingly, the authors introduce Emotions information into the GA's status, emphasizing their role in human-like behavior. To effectively simulate real-world scenarios, agents must have cognition, attitudes, and personalities. However, human decisions are often emotionally driven, shaping their choices and interactions. Hence, emulating such aspect in GAs is crucial for realism, as it influences how they express intentions. While modeling emotions is challenging, LLMs' rich behavioral knowledge offers a way to enhance GAs, making simulations more lifelike [52], [53], [54].

2) *Towards Urban GABMs*: Despite the novelty of GABMs, ongoing research explores their potential, particularly in urban simulation. While fully validated applications remain scarce

in top-tier publications, conceptual frameworks highlight their potential real-world impact.

In urban public transportation, Liu et al. [55] discusses how GABMs model commuter behavior in response to policy shifts, infrastructure changes, or disruptions. Unlike traditional ABMs, which rely on predefined rules, LLM-powered agents dynamically adapt using real-time data, such as social media updates and policy announcements. This flexibility enables planners to test counterfactual scenarios and refine strategies, particularly in large cities facing unexpected events like extreme weather or labor strikes.

Beyond transit, [56] presents a GABM platform for modeling socioeconomic neighborhood evolution by simulating interactions among individuals with varying economic backgrounds and mobility patterns. GAs process socio-political discourse, housing trends, and community sentiments to assess the impact of policies like affordable housing programs or zoning laws. Compared to conventional agents, GAs better represent diverse personalities and autonomously generate negotiation and decision-making behaviors, reducing the need for complex manual specifications and enhancing simulation realism.

V. INFORMAL THEORY OF ABMS VALIDATION

In ABMs, validation is essential for gaining trust and credibility in simulation models, particularly ensuring that the model meets its intended purposes and generalizes to real-world social dynamics. In this field, validation is an intricate problem because of the complex mechanisms and scenarios these models attempt to represent. Moreover, there is a lack of universally accepted standards for ABM validation, which makes it even more difficult for model designers to demonstrate the validity of their system. This introduces the need for formal methods and rigor in the validation processes for ABMs [57], [58].

Furthermore, an explicit hierarchy of evidence should guide ABM validation. This can be inspired by the one used in evidence-based medicine and policy-making, where the strongest evidence comes from ecological validity and rigorous experiments, while observational data and consistency with previous theories are considered weaker [59]. It is worth noting that the rigor required for validation depends, again, on the purpose of the ABM. In a model intended only to guide further research, the threshold for evidence can be reasonably lower. It's crucial to note that the model's hypothesis should not only be evaluated based on direct evidence but also on their "productivity" or their ability to stimulate further empirical research and theoretical developments [60]. On the other hand, when the model is used to guide real-world decisions with significant consequences, validation should require higher rigor and evidence.

A. Foundational and Advanced Methods

A recent attempt to define guidelines for ABM validation has been proposed in [61], where the authors, after reviewing different validation approaches, discuss how the choice of validation methods should be tailored to the specific needs of the simulation, the characteristics of the model, and the available data. In particular, they classify nine key validation-supporting

methods into foundational and advanced. Such a classification follows, along with a brief description of each method.

1) Foundational Methods:

- a) *Data analytics*: Essential for handling input and output data, data analytics supports the validation by ensuring data integrity and relevance. This method involves extensive data cleaning, organization, and analysis, including statistical tools.
- b) *Docking*: A method for comparing the outputs of a simulation with the ones of other state-of-the-art ABMs to verify if they align with each other. It is impractical if no appropriate reference model is available, as in the case of novel simulation studies.
- c) *Empirical validation*: Involves using real-world data to ensure the model's predictions are realistic and grounded in actual data. Particularly useful when historical or observational data is available, however, it is less effective in cases where sufficient empirical data does not exist.
- d) *Sampling*: Covers the various sampling techniques used to explore the simulation space and understand the impact of different input variables (e.g. decision thresholds). It may require repeating simulations and hence, extensive computational resources.
- e) *Visualization*: Focuses on using graphical representations to understand and analyze the behavior and outcomes of simulations. Yet, its reliance on qualitative interpretation can lead to subjective biases.

2) Advanced Methods:

- a) *Bootstrapping*: It addresses issues of limited real-world data by generating empirical distributions of the model's output through resampling methods. It can highlight robustness of the model results, but it may be computationally intensive.
- b) *Causal analysis*: Look at the model's causal relationships to identify and correct any erroneous or unexpected behavior. It demands rigorous statistical modeling and well-structured rules from domain knowledge.
- c) *Inverse generative social science*: It uses techniques like evolutionary algorithms to explore various potential agent behaviors and their consequences within the model (e.g., by using back-tracking of the actions). However, its reliance on predefined behavioral constraints may limit its generalizability.

Each validation method has distinct advantages and limitations, making their applicability highly dependent on the model's purpose, data availability, and computational constraints. Indeed, one can think of a cohesive validation framework, which combines multiple methods to address different aspects of validation within a single ABM. For example, a combination of empirical validation, visualization, and sampling can be used in an agent-based model designed to study urban traffic flow. *Empirical validation* can ensure that simulated traffic patterns align with real-world traffic data. At the same time, through *Visualization* researchers can observe the flow of vehicles and detect potential anomalies in emergent congestion patterns. Simultaneously, sensitivity analysis through *Sampling* methods can help identify how small changes in driver behavior or infrastructure

modifications impact overall traffic efficiency. Additionally, *Causal Analysis* can be employed to determine whether unexpected traffic bottlenecks emerge due to model assumptions or real-world behavioral factors.

B. Empirical and Face-Validation

The idea of tailoring the validation methods to the specific needs and purposes of the simulations is further supported by previous theory on ABMs validation [58]. In this work, the authors provided guidelines for rigorous validation—meaning the model corresponds to reality. According to them, there are four steps for considering a model to be rigorously validated.

Two of these steps focus on *empirical validation* of ABMs, specifically concerning the validation of the model’s input (step 1) and output (step 2). Using statistical measures, they ensure that these elements align with real-world data or are comparable to other models’ outcomes. For example, in traffic flow simulations (e.g., in Table I), traffic patterns must be compared to actual traffic data. This is required to evaluate the model’s accuracy when simulating the real world. These validation steps can be easily performed by using the *Foundational Methods* previously described. However, they require the availability of data or the existence of a similar model for comparison.

The remaining two steps from [58] involve the concept of “*face*” validation, and they do not necessarily rely on data availability. Rather, they rely on human intelligence through expert assessments and structured walk-throughs. They assess the model processes and outcomes appear reasonable and plausible w.r.t. expert and stakeholder knowledge. Such validation steps are namely

- 1) *Microface validation*: To validate that the individual-level mechanisms and properties of the model correspond to the real-world ones. It checks if the agents and their actions realistically reflect human-like dynamics.
- 2) *Macroface validation*: To validate that the aggregate patterns and dynamics of the model correspond to real-world phenomena. The focus is on the emergent properties and overall behavior of the system.

Back to the traffic flow example, in a microface validation scenario, experts should be involved in validating a single agent’s directions. In contrast, in a macroface validation, they should focus on validating traffic congestion dynamics at an aggregate level.

In both micro and macrosenarios, the model is not directly compared to real-world data. Instead, the emphasis is on showing that the general model’s attributes and processes reasonably correspond to the ones of the real world.

Microface validation is particularly relevant because it allows us to examine the system’s most detailed and specific aspects (such as the individual behavior of GAs).

Conversely, macroface validation is essential because it offers a broader perspective of the system. Such approaches are particularly useful when empirical validation is not possible due to the absence of data.

1) *Face-Validation: HowTo*: According to [62], *face*-validation involves three key methodological elements.

- a) *Animation assessment*: Experts review simulation animations to ensure the system’s or its components’ behavior is realistic. Visualization methods can help observe the simulation from a general perspective to assess dynamics like passenger flows or traffic jams and potentially track individual agents to scrutinize their specific behaviors.
- b) *Output assessment*: This involves human experts examining the simulation outputs for plausibility. This includes checking whether their relationships and the dynamics and trends observed in the simulation outputs are coherent. This assessment can be automated if the relationships among the inputs, the evolution, and the outputs can be formally defined (e.g., through constraints and rules definition).
- c) *Immersive assessment*: Experts immerse themselves in the simulation by adopting the perspective of an individual agent within the system. This allows them to directly evaluate the believability of an agent’s behavior and reactions, as well as the interactions and reactions of other agents within the simulation environment. The effectiveness of this method depends significantly on the design of the user interface, which may need to be customized for different types of agents.

Animation and immersive assessments share the fundamental principle wherein human validators replay simulations to examine the phenomena within them. In the immersive assessment process, validators impersonate agents and make informed decisions that are subsequently compared to those made by the simulated agents, providing a more advanced framework that allows validators to experience the simulation and have a complete sensory perception fully, for example, through virtual reality [63]. This approach is particularly valuable when validating human emotions is necessary, but relevant data is unavailable.

Overall, these validation techniques help ensure that the model is not only statistically and structurally sound but also intuitively plausible to human experts, crucial for its acceptance and usefulness in real-world applications.

VI. GABMS VALIDATION

All the aspects and concerns about traditional ABM validation are inherited by GABMs, although the latter introduces additional unique challenges. For example, whether or not the simulation involves GAs, ABM’s *foundational* and *advanced* validation methods remain useful. However, GABMs involve unique challenges, such as ensuring believable behaviors of GAs and how social dynamics correlate with GAs’ evolving statuses or conversations. For instance, as also discussed in [3], the context of urban simulation requires checking the simulated residents’ behaviors, information diffusion, and consistency according to the conversations between GAs, the history of the simulation, and the surrounding environment. Moreover, it is worth noting that when discussing GAs performing actions or visiting places, we do not mean these agents have human-like abilities or intentions. Their actions are designed to seem realistic and engaging, but this does not imply they possess

true independence or consciousness. Indeed, current state-of-the-art GAs rely on LLMs to enhance the illusion of human-like behavior, as this new generation of language models encapsulates information on human behavioral patterns. However, this is only believable to a certain extent due to issues such as hallucinations and biases in LLMs, introducing new validation challenges. Additionally, there is a lack of automatic method for validating LLMs, and the current trend involves employing a large number of human validators to assess these models, followed by a voting system to compute performance metrics based on human evaluations [64], [65].

Recent studies [66], [67] discuss how current LLMs lack cognitive competence in ways that sometimes produce fragile and even bizarre behavior, highlighting limitations model's cognitive abilities, context sensitivity and causal reasoning. However, the study demonstrated that GPT-3 could solve many tasks reasonably and showed some human-like behaviors. GAs inherit the mentioned limitation when used in GABMs, and they become even more severe in the case of conversational agents, where the validation can require the analysis of natural language interactions. Again, human validators can be employed in this case. Still, it is easy to understand that this approach does not scale to simulation involving thousands of interacting agents (e.g., in cities simulations).

As of today, while there exists a wealth of literature on ABM validation methods, it is challenging to find new specific validation methods proposed for GABMs. Therefore, we leverage ABM literature—particularly concerning *face-validation*—as a basis to discuss how the previous theory can be applied to the transition to GABMs:

- 1) *Microface validation*: When transitioning from traditional agents to GAs, two dimensions of this type of validation emerge: the validation of individual decision-making processes and the natural language interaction. For the decision-making processes, the validation should focus on the decisions made by GAs and assess the believability of social dynamics. For interaction, it should examine the conversations between agents at a deeper level of granularity to ensure that decisions are coherent with the GAs' natural language interaction history. Additionally, validating emotions and other qualitative aspects, such as the consistency in the agents' personality, can be seen as a third dimension.
- 2) *Macroface validation*: While this approach does not significantly differ from traditional ABM to GABM, it is essential because it enables the validation of systems where individual agent validation is impractical (e.g., simulations with thousands of agents). Advanced validation methods such as bootstrapping, causal analysis, and inverse generative social science prove invaluable.

Face-validation offers a framework for verifying systems without empirical data. With LLMs, we can simulate scenarios using embedded knowledge, enabling populations to generalize across locations without real-world data. Leveraging face validation is key to maximizing this potential.

In this context, promising approaches can be

- 1) *Survey-based validation*: Interview GAs regarding qualitative aspects, decisions and historical coherence. Responses from GAs are later analyzed by humans (e.g., in [3]).
- 2) *Observational studies*: Analyze human interactions across different settings and compare them with agent behaviors to ensure consistency and realism in simulations (e.g., in [46]).
- 3) *Expert reviews*: Engaging experts of different domains to review the agents' behavior, providing scores on their plausibility. Here, experts do interact with GAs; rather, they check agents' behaviors from an external point of view.

It is essential to note that such methods could serve as a baseline for validation, but specific and general solutions should be developed on top of them. Combining techniques can generate a more comprehensive approach to GABM validation.

In Fig. 3, we present a cohesive framework designed to facilitate understanding of this section. This framework illustrates how the validation of GABMs can leverage established validation methods used for traditional ABMs, while also addressing the specific validation requirements of GAs. To achieve this, traditional approaches can be integrated with specialized validation techniques tailored for GAs, enabling a comprehensive validation strategy for different validation purposes.

Currently, there is no agreed-upon method for interpreting the outcomes of simulations that use LLMs to model human populations. The critical question remains: What criteria should we use to determine if, how, and under what conditions the findings from simulations might apply in real-world settings? Addressing this challenge requires collective and multidisciplinary effort from the research community. While the debate continues, the priority should be developing formal methods to ensure model outcomes can reliably generalize to real-world dynamics [68].

A. Existing Gas—Validation

This Section details how the developers of state-of-the-art GAs (discussed in Section IV) designed validation processes to assess the effectiveness and realism of their systems. A comparative summary of the following validation approaches for GABMs is provided in Table II.

As a first example, developers of [47] use visualization, data analysis, and sampling methods to validate their simulations. Interestingly, this demonstrates how, in the case of inferential GAs, GABMs can be validated even only employing one or a set of foundational methods for traditional ABMs that are widely accepted by the community (see Section V).

Delving into conversational agents and in line with what we mentioned about leveraging human validators, authors of [3] recruited 100 paid evaluators for the validation process to demonstrate the effectiveness of the proposed architecture in generating believable behaviors. The validation process encompasses two phases. The first one is a *Controlled* evaluation, and the second one is an *End-to-End* evaluation.

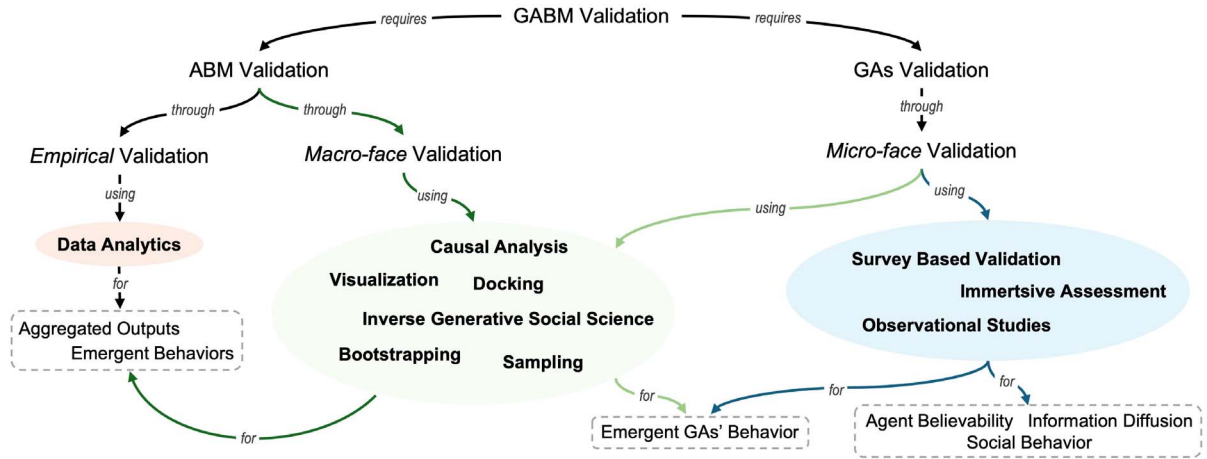


Fig. 3. Validation workflows for GABMs, illustrating how various validation methods align with distinct validation purposes. Methods discussed in Sections V and VI are shown in *Italics*, while validation objectives are enclosed in a dashed box. The framework shows how GABM validation integrates traditional ABM validation approaches with specialized techniques for validating GAs, for *empirical*, *macroface*, and *microface* validation pathways.

The *Controlled* evaluation seeks to answer the question: “*Do individual agents properly retrieve past experiences and generate believable plans, reactions, and thoughts that shape their behavior?*”. For this purpose, human validators interviewed the agent on five question categories (self-knowledge, memory, plans, reactions, and reflections). They use *believability* as an evaluation metric, measuring it according to human evaluators watching the replay of a 2-day simulation of a randomly sampled subset of agents. The system ran for five different conditions (ablation), and the evaluators ranked the conditions as most or least believable.

On the other hand, with the *End-To-End* evaluation, the authors aim to answer the question: “*Does a community of agents demonstrate information diffusion, relationship formation, and agent coordination across different pockets of the community?*”. For this purpose, they run a 2-day simulation by tracking two pieces of information from two single agents: the organization of a party and the candidacy for mayor. They test for: information diffusion by interviewing the 25 agents with “Do you know about the party?” and “Do you know about the candidacy?”; relationship information by constructing an undirected graph of agents where edges represent mutual knowledge between two of them, and they measure the density of the network (increases as the simulation keep going); agent coordination by counting the agent which actually went to the party.

While they designed a state-of-the-art architecture for GAs, overcoming the limits of existing GAs whose believability falls short in extended simulation, they validated their system using a qualitative measure of coherence between the agents’ behavior and the expected behavior of humans in similar scenarios, often influenced by historical context, rather than a formal representation of believability. Their method recalls the *microface* validation approach at a conversational level and the concept of *Immersive Assessment* discussed in Section V, where human validators assess the agents’ behavior by experiencing the evolving simulation. Indeed, in this particular case, the

immersive experience engages validators in the conversation between agents and their resulting decisions.

As previously mentioned, in [3] authors involved 100 human validators to inspect the behavior of 25 agents. Inspired by this approach, also in [37], the authors validate their simulations involving 30 human participants to verify the behavior of 10 GAs. This suggests that for GABMs, validation requires more human evaluators as the number of artificial agents increases, making large- and small-scale validation challenging without sufficient contributors. Additionally, expert-designed interviews introduce subjectivity, making the process qualitative rather than formal.

In [15], the authors, rather than proposing a concrete validation approach, discuss best practices for GABMs validation. The validation framework that the authors discuss is based on Empirical Validation, referred to as the *Similarity Principle*. This principle emphasizes the similarity between tested and untested samples. A model makes a family of related predictions, and testing one prediction can provide confidence in similar, untested predictions. In the context of human-behavior validation, the authors advocate that the gold standard is the direct measurement of model predictions on new test data that could not have influenced the model’s parameters. For example, if a model predicts how humans will behave in a particular situation, the best evidence is measuring actual human behavior. The authors emphasize this approach as the most robust form of validation. While true, we underscore how this would require availability of data or the development of frameworks for immersive validation, tailored to the specific GABM. Authors also provide validation recommendations that extend beyond empirical testing by discussing the concept of *Algorithmic Fidelity*. Algorithmic fidelity describes how well a model can be conditioned using socio-demographic backstories to simulate specific human groups. This fidelity must be measured anew for each research question, as it may vary across different topics and experiences. One direction for further research in this context would be based on the work by [69], which involves checking

for the fidelity of human personality traits in LLMs. As a concrete validation practice, it is essential to check models for consistency with prior theory, whether they pertain to psychological aspects (e.g., testing human behavior) or structural elements of the real world (e.g., validating a GABM modeling consumer behavior by showing that prices in the model move according to classic microeconomic theories). Additionally, the authors in [15] discuss open challenges and unresolved issues concerning *train-test contamination*, *representation of stereotypes*, and *detail limitation*. Finally, they emphasize the importance of addressing validation of *Hard-to-Reach Populations* GABMs, where individuals are difficult to recruit for experiments.

In the work by [46], the authors used *empirical validation* for different aspects of their system, such as *information propagation*: the system simulates the temporal dissemination of events (e.g., eight-child mother and the Japan nuclear wastewater release events). It compares these simulations with real-world data to assess model accuracy in forecasting propagation patterns, capturing distinct emotional peaks and dissemination trends; *emotions and their propagation*: emotions are classified into three levels: calm, moderate, and intense. The model predicts the user's emotion level in subsequent time steps by considering the current emotion level, user profiles, user history, and messages received. Again, the effectiveness of the model is demonstrated by its ability to reproduce the dynamic process of emotion propagation observed in real-world data; *interactive behavior*: The model simulates users' decision-making processes regarding whether to forward content, create new content, or remain inactive. Validation is performed through metrics like accuracy, AUC, and F1-score against actual data.

While the authors gathered real-world labels from social networks, in general, collecting emotional and qualitative data across scenarios remains challenging.

A promising approach to GAs validation has been proposed by [4], where the authors introduce a comprehensive framework for evaluating agent interactions across various scenarios. Interestingly, they designed a holistic social agent evaluation framework, SOTOPIA-EVAL, using seven dimensions grounded in established literature from sociology, psychology, and economics. Both human annotators and GPT-4 are used to evaluate agents based on this set of dimensions: *goal completion* to consider whether the agent met its explicit goals; *believability* as the assessments of realism and consistency (alignment with the agent's personality, values, and background); *knowledge* gained, its relevance, and importance to the agent's goals; *secrets* the agent successfully kept and the impact of any leaks; *relationship* to measures whether the interaction has a positive or negative effect, including changes in the perceived quality of the relationship, social status, and reputation resulting from the interactions; *social rules* to assess whether the agent violated any social norms or legal rules; *financial and material benefits* as short-term monetary gains, long-term economic benefits, and overall financial outcomes of the interactions.

This comprehensive framework for evaluating agents is holistic, yet it misses critical qualitative dimensions such as emotions, mood, or attitude [46], [70], [71].

Notably, SOTOPIA-EVAL integrates an LLM (GPT-4) to automatically evaluate the mentioned dimensions. In particular, authors instructed both human and GPT-4 validators with the same questions and metrics to assess GAs interactions, enabling them to assign numerical scores. Results show that GPT-4 aligns well with human evaluators in structured aspects like goal completion, financial benefits, and relationship preservation. However, it struggles with more nuanced evaluations, particularly social norms and secret-keeping. Hybrid validation, combining LLM assessments with human oversight, remain a promising approach for context-aware evaluations of GAs.

VII. ABM VS GABM: A COMPARISON

After examining the technical aspects, applications, and validation of GABMs, we now summarize their advantages, limitations, and key challenges. In this section, we explore the advantages of ABMs over GABMs and vice versa, highlighting the strengths and limitations of each approach across different dimensions. We present a comparative analysis, summarizing key insights in Table III.

With the following discussion, we want to highlight how ABMs and GABMs are not mutually exclusive but complementary, each addressing gaps the other struggles to fill.

A. Why ABM

When comparing ABMs to GABMs, where LLMs power agents, traditional ABMs offer several advantages in terms of computational efficiency, interpretability, control, and scalability. By referring to Table III, in this section, we discuss the advantages of ABMs against GABMs across different dimensions (in bold).

First, we shed light on the *Resources* needed to design and run the simulations. Traditional rule-based ABMs use simplified agents with predefined behaviors, making them significantly more computationally efficient than LLM-powered GAs, which require extensive processing power to generate responses. This is also the case with EA- and LR-based approaches since the first may require multiple generations of agents to evolve through mutation and crossover, and the second must explore and optimize its behavior over time.

Furthermore, rule-based ABMs are highly scalable because simple agent rules can be efficiently simulated for thousands or millions of agents with minimal computational overhead. This is not the case with RL-based ABMs and GABMs, which require significantly more memory and processing power per GAs, making large-scale simulations impractical or extremely costly. On the other hand, EA-based approaches generally scale better than RL in ABMs, as the evolutionary process allows populations of agents to improve collectively without requiring time-consuming exploration strategies.

For what concerns *Agent Design*, when realizing traditional ABMs, developers typically have precise control over agents'

TABLE III
COMPARISON BETWEEN ABM AND GABM

	ABM			GABM
	Rule-based	EA-based	RL-based	
Resources				
Computational Efficiency				
Scalability				
Data or Domain Specific Knowledge				
Agent Design				
Control over Agents				
Learning During Simulation				
Real-time optimization				
Hyperparameters Tuning				
Fast Prototyping				
Agent Behavior				
Determinism				
Reliable Decision-Making				
Evolution				
Cognitive Functions Simulation				
Language-based Human-like Interactions				
Flexible and Reactive Decision-Making				
Nuanced Mental Models				
Validation				
Transparency of Decision-Making processes				
Efficiency				
Direct Human-Agent Interaction				

Note: A green cell indicates that the row item represents an advantage for the corresponding approach. Conversely, a red cell indicates that the row item represents a limitation.

behavior, which can be explicitly devoted to fit domain-specific needs. With LLM-powered agents, behaviors emerge from pre-trained models, making it harder to specialize specific agent interactions without extensive prompt engineering (e.g., using RAG [72]) or model retraining. Moreover, allowing agents to learn from interactions and the evolving environment during simulation (as in AE- and RL-based) is unpractical for GAs merely relying on LLMs.

Development practice and control in traditional ABMs guarantee stability and predictability of *Agent Behavior*, meaning that rule-based ABMs produce consistent and replicable results because they rely on deterministic models that ensure agents always act according to the defined logic. In contrast, EA-based ABMs introduce randomness through genetic variation, leading to nondeterministic outcomes. Alongside them, RL-based approaches introduce stochasticity in agents' decision-making, making agent behaviors less deterministic due to inconsistent results during early training, when faced with unexpected scenarios or trapped in suboptimal solutions during optimization. Also GAs can introduce nondeterminism and variability due to the probabilistic nature of LLMs, potentially leading to inconsistent outputs across multiple runs of the same simulation. Moreover, being based on LLMs, these agents may hallucinate or produce unrealistic, inconsistent, or contextually irrelevant responses.

Finally, when it comes to *Validation*, explicit rules and well-defined behavioral mechanisms in ABMs make agents' decision-making processes fully transparent, interpretable, and efficient. Also, EA-based approaches offer moderate transparency, as evolved behaviors can be back-traced to selection and genetic mutations. In contrast, GAs and RL-based

agents function as black boxes, making it difficult to trace why a particular decision or interaction occurred. This lack of transparency poses significant challenges in the validation of GABMs, often requiring substantial resource consumption—both computational and human-based (see Section VI-A). In this context, the validation of RL-based agents—which do not involve natural language interactions—is more efficient w.r.t. GAs validation, as it can be performed through traditional performance metrics and reward optimization.

Overall, traditional ABMs remain superior in efficiency, scalability, control, and interpretability. Depending on the application, the deterministic and computationally-light nature of traditional ABMs often makes them a more practical choice.

B. Why GABM

GABMs offer significant advantages over traditional ABMs, particularly in terms of adaptability, cognitive reasoning, and interaction complexity. These advantages make them well-suited for complex, dynamic, and multidomain environments where traditional ABMs may struggle. Referring to Table III, we explore the unique advantages GABMs introduce compared to traditional ABMs.

While GABMs are less efficient in terms of computational performance and scalability compared to ABMs, they provide distinct advantages concerning *Resources* requirements and *Agent Design*. Specifically, the integration of GAs enables data-less simulations, allowing for greater flexibility in prototyping while simplifying model development and adaptation across different domains. Indeed, traditional ABMs require extensive domain-specific data, knowledge, and manually defined rules to operate effectively—whether rule-based, EA- or

RL-based. GABMs with LLM-powered agents, in contrast, can generalize across different domains using pretrained knowledge from diverse datasets, making them useful for scenarios with incomplete or uncertain data. GABMs allow for rapid prototyping and experimentation, as the agents can generate adaptive responses without requiring highly specialized programming.

Promising advantages emerge from the *Agent Behavior* perspective, as recent experiments have demonstrated how LLMs can encapsulate human behavior [3], leveraging the vast amount of data they are trained on. This allows for enhanced cognitive function simulation and enables more flexible and reactive decision-making. Unlike traditional ABMs, GAs can adapt dynamically to changing environments, exhibiting context-aware behaviors that allow them to reason and react in real-time rather than following predefined decision rules. GABMs introduce, moreover, the opportunity to simulate language-based human-like interactions, where agents can engage in rich exchanges with other agents, emulating negotiations, multiagent collaborations, and decision-making in ways that traditional ABMs struggle to replicate. Furthermore, LLMs incorporate nuanced mental models, enabling GAs to simulate beliefs, emotions, and intentions, leading to more cognitively driven behaviors.

It is worth noting that GABMs are intrinsically evolutionary, as they make decisions based on the evolving history of the simulation. Unlike RL and other EA approaches, they do not require real-time optimization during the simulation, thereby avoiding startup imprecision and allowing agents to function effectively from the outset. Since GAs produce fully evolved agents beforehand, GABMs can preserve the evolutionary nature of the simulation without the need for continuous optimization and hyperparameter tuning, but relying on the processing of *Agent Status* through LLMs (see Section IV).

Finally, even though *Validation* of GABMs introduces significantly more complexity than traditional ABMs, we can identify an unprecedented advantage that the presence of GAs introduces in validation processes: unlike traditional ABMs, LLM-powered GAs can interact directly with human users, enabling real-time validation and experimentation in complex social environments. In this scenario, developers and human validators can directly interview the agents—during or at the end of the simulation—to study the correlation between individual decisions (or social dynamics) and agents’ cognitive functions.

GABMs with LLM-powered agents can provide more realistic human-like interactions, cognitive simulations, and multidomain flexibility, without the need for real-time optimization and costly hyperparameters tuning. These properties make GABMs particularly advantageous for social simulations where rule-based approaches may be too rigid or simplistic. However, their higher computational cost, unpredictability, and validation remain key challenges. Combining the strengths of ABM and GABM offers a promising way to overcome their limitations, a topic we will explore further in the discussion.

VIII. GABMS: EMERGING CHALLENGES

While the advent of GAs in ABMs presents significant opportunities, several challenges remain. A key issue stems

from the underlying technology that powers LLMs and inherently GAs within GABMs: LLMs’ hallucinations. This adds a layer of complexity that extends beyond the validation of GABMs as a whole, requiring assessment of the individual behavior of GAs—which is unnecessary when agents are rule-based in traditional ABMs. Another critical challenge involves the computational resources required for large-scale simulations, where thousands of GAs dynamically interact.

A. LLMs’ Hallucinations

Hallucinations in LLMs refer to generating factually incorrect or ungrounded outputs, categorized into factuality and faithfulness hallucinations [73]. Factuality hallucinations involve contradictions to real-world facts (e.g., Edison inventing the telephone) or fabricating unverifiable information. Faithfulness Hallucinations occur when outputs deviate from input, misalign with context, or exhibit logical inconsistencies. These errors arise from the probabilistic nature of LLMs, which generate text based on token predictions rather than fact retrieval. Causes span data, training, and inference stages. Data-related issues stem from misinformation or biases in pretraining corpora, and can be mitigated via data filtering and retrieval-augmented generation (RAG) [72]. Training-related issues include limitations in fine-tuning and misalignment, often addressed through strategies like sensitive neuron dropout [74] and RLHF, a technique which realign responses using human feedback and RL [75]. Inference-related issues involve decoding flaws, overconfidence, and reasoning failures, often controlled via constrained decoding [76] and multiagent verification [77].

In GABMs, hallucinations further amplify the need for validation, as they can undermine believability and produce incoherent agent behaviors.

B. Validation

New aspects of simulation arise with the interplay between qualitative and quantitative elements, like agents’ interpretations of their environment, agents’ social interactions, consistency in personalities, and the effect of emotions. Validating these aspects is difficult due to limited data, requiring alternative validation methods. Following face validation principles (see Section V), researchers should develop advanced approaches to ensure the reliability of GABMs, laying the foundation for the development of the field.

We advocate that the research community in this field should focus on new criteria for validating GABMs and the conditions under which GAs can be used as a proxy for human behavior. For this, it is essential to define validation methods that confidently detect if a model correctly represents the real world, using approaches with slight dependency on data to ensure their broad applicability in different contexts. It is, moreover, crucial to consider validation methods from various disciplinary perspectives. These perspectives may be independent of the specific application field of the simulation but can depend on the human-like behavior we aim to reproduce. Qualitative interpretation and validation are essential in place of or alongside quantitative ones.

Beyond the validation methods requiring human intervention, recent advancements provide new avenues for validating GABMs, mainly through the use of automated validation strategies. One promising approach is leveraging LLMs themselves as validators to assess the coherence of agents' behavior. For example, frameworks such as SOTOPIA [4] demonstrate how LLMs can systematically evaluate social interactions, goal completion, and behavioral consistency. Grounded on the fact that GAs can entertain natural language conversation, they provide an interview mechanism. With this mechanism, LLMs-evaluators assess GA's behaviors through natural language-based interviews. Being an automated approach, the field of GABMs can benefit from more scalable and context-aware validation that aligns with human judgment and social dynamics. With this purpose, specific fine-tuning strategies on LLM validators can be applied to specialize them in social interaction dynamics [78], making them able to assess qualitative aspects such as personality coherence, logical consistency in agent decisions, and adherence to social norms. Additionally, automated validation pipelines can integrate self-reflective mechanisms where GAs evaluate their own outputs for contradictions or implausibilities during the simulation.

However, while these methods reduce the reliance on empirical data, they require careful calibration to mitigate biases inherent in LLMs. Future research should focus on refining LLM-based validation approaches by combining them with expert review and observational studies.

C. Computational Costs

In the most prominent studies on GAs, the authors remark on the need for prohibitive computational capacity, since GAs require a large amount LLMs' generations for each interacting agent. Unlike traditional ABMs with simple, low-cost rules ($O(1)$ or $O(N)$ for N agents), LLM-driven agents introduce nondeterministic and dynamic computational paths, making worst-case complexity harder to bound. The computational cost of GAs is mainly driven by LLM inference complexity, multi-agent scaling, and memory retrieval. LLMs rely on attention mechanism with quadratic complexity, $O(n^2 \cdot d)$, with n number of tokens and d hidden dimension [79]. Moreover, as the number of GAs grows, interaction complexity scales nonlinearly. Memory retrieval also adds overhead, with vector search mechanisms scaling between $O(\log N)$ and $O(N)$, depending on indexing strategies [3].

From a resource requirements perspective, although scaling GABMs can be challenging and seems unfeasible, it is essential to note that the multiple agents generations and interactions are parallelizable. Additionally, applying model compression techniques (such as memory summary in [3]) can contribute to significant costs reduction. Hybrid approaches, using smaller LLMs or rules for noncritical agent decisions while reserving larger models for high-impact interactions could also enhance cost efficiency.

Although these practices can reduce simulation costs, developers must critically assess whether the added complexity of GABMs justifies the interpretative benefits. Simplified

models may suffice, whereas highly detailed simulations risk high computational costs with minimal accuracy gains. In practice, developers should be able to assess the cost-benefit trade-off between a GABM's usefulness and its costs before running a simulation. The lack of precise data and benchmarks on resource requirements, introduce new research avenues for developing predictive cost models and adaptive resource allocation strategies to bridge this gap.

IX. DISCUSSION

A. Addressing Broader Questions

We further explore GABM potential to address fundamental questions that drive ABM research and remain challenging to tackle with traditional approaches alone.

The first fundamental question in ABM is: *how macroscopic patterns emerge from microscopic interactions?* The introduction of GAs contributes by simulating diverse and dynamic behaviors, allowing for more realistic and varied interaction patterns. This capability extends beyond simple rule-based agents by incorporating creativity, personality-driven relationships, and nuanced decision-making.

Building on this, another key challenge is understanding: *how new ideas and information spread through populations?* Traditional ABMs model diffusion through network structures, but they often rely on static or probability-based adoption rules [80]. GAs, in contrast, simulate the emergence, adoption, and dissemination of innovations dynamically, embodying traits like curiosity, risk-taking, and adaptive reasoning. Studies, such as [3], investigate how information spreads over a network of living GAs by analyzing their memory, hence extracting perception-based behavioral patterns (see Section VI-A).

Adaptation is another crucial aspect of complex systems, leading to the question: *how do agents modify their strategies in dynamic environments?* Traditional ABMs often rely on static behavioral rules or objective function to optimize, whereas GABM allows agents to evolve autonomously. This makes it possible to explore how cooperation, competition, and trust emerge in resource-limited environments, where adaptive behavior plays a critical role. Pioneering works use GAs in game settings [19], [81] emphasizing the importance of spontaneous phenomena, wherein GAs deeply engage in contexts and make adaptive decisions without explicit directions.

In this context, it is natural to ask: *how social networks and interactions influence collective behavior?* GAs can better model realistic social behaviors, incorporating factors like persuasion, conformity, and rebellion, which are critical for studying phenomena such as misinformation spread, echo chambers, and online polarization. A classic example is Schelling's [13] segregation model, which, despite its well-known explanatory power, cannot capture the full spectrum of human biases and motivations. While GABM introduces greater behavioral complexity, it also allows for richer analyses of how individuals' belief systems and interactions shape larger societal trends, from segregation and polarization to consensus formation. The

authors of [46], for instance, investigated opinion diffusion in an online social network of interacting GAs, accurately predicting polarization patterns.

Lastly, GABMs expand the study of social simulations and human behavior, addressing complex dynamics beyond traditional ABMs, even those with evolutionary adaptations. They allow exploration of profound, singular questions such as: *How do linguistic and cognitive biases influence social dynamics in agent-based models? Can GAs develop emergent beliefs or misinformation without explicit training?* Unlike conventional approaches, GABMs introduce a contextually adaptive framework where agents interact in ways more closely aligned with human social processes, potentially providing us with the tools to answer such new questions.

1) *From Quantitative to Qualitative Interpretation:* GAs unlock the potential for qualitative interpretations of humans-related simulations, addressing a notable limitation of previous approaches [82]. By incorporating models of human cognition and using LLMs, agents can exhibit cognitively plausible behavior consistent with human decision-making processes, including emotional responses and the influence of incomplete information [71], [83]. This could help understand complex social ecosystems previously studied through surveys and long-term monitoring of the actual population of interest. In this context, GABMs enable the exploration of a qualitative dimension of simulations by accounting for how agents' behaviors might be influenced by, for example, their emotional states. For instance, let us consider a traffic flow simulation: in a traditional ABM, route selection is typically driven by rational factors, such as choosing the shortest path between two points. However, in a more realistic scenario, human decision-making is often influenced by emotions, personality, attitude, and memories. An individual might choose a longer route to avoid a location that evokes fear or discomfort, or they might select a path for its scenic value, even if it is less efficient.

From a traditional ABM point of view, we can consider these as fluctuations in the agents' decision-making, treating them as noise. Or rather, one can think these "anomalies" are well represented and implicitly embedded into real-world data.

Although these are valid interpretations, real-world data are not always available, especially when we want to simulate scenarios of emergency or when we want to study the social dynamics in less developed areas (e.g., in informal settlements or unfriendly territories) where technology falls short. More interestingly, GABMs can be used to enrich simulations, making it possible to uncover key aspects of social dynamics related to emotions, such as:

- a) *Emotional triggers:* Since the agents react to events based on emotional states, we can look deeper into how emotions affect human behavior and decision-making.
- b) *Qualitative analysis:* We can study how different intervention and simulation parameters affect emotions and human-like behavior.

Such aspects are difficult to capture with traditional ABMs, for which they typically require data availability domain specialized design.

B. Cities: A Natural Testbed for GABMs

When we talk about social phenomena, we often refer to interactions that take place in the context of communities, whether these are sets of things, animals, or human beings. In the latter case, urban communities represent a highly dynamic environment where human interactions shape complex social, economic, and infrastructural systems. Given their structured yet evolving nature, cities provide a fertile ground for deploying GAs.

Since LLMs are trained on extensive corpora of human text, they inherently capture social norms, moral values, and behavioral patterns as evidenced by [38], [84]. This enables GAs powered by LLMs to simulate realistic social behaviors, decision-making processes, and interactions within digital urban environments. Notably, the seminal work on GAs [3] has been explored in Smallville, a virtual city, where autonomous AI-driven agents could exhibit emergent social behaviors such as forming relationships, exchanging information, and collaborating in shared tasks.

Beyond urban planning and transportation, studies have also demonstrated the emergence of complex social behaviors in agent-based simulations, where LLM-driven agents exhibit human-like cooperation, negotiation, and adaptation in multiagent environments [37]. These findings further validate that urban communities, characterized by their structured social interactions and evolving dynamics, are a natural testbed and a crucial application domain for GAs. Hence, GAs could have a transformative impact on urban planning and policy-making. By representing residents' perceptions of urban spaces, we move beyond simplistic models to a more nuanced understanding of urban environments. Indeed, the qualitative interpretation facilitated by these models aims to comprehend subjective experiences, feelings, and the underlying meanings behind actions, of great importance in urban dynamics [85]. This approach enables urban designers to create detailed experience maps, visualizing the emotional highs and lows that users experience when engaging with various urban spaces [86]. Such qualitative interpretation, unpracticable with traditional approaches, unlock unprecedented opportunities for improving the overall urban experience, leading to more responsive and human-centered urban planning. This approach promises to transform urban planning from a primarily functional exercise to one that truly enhances the quality of life for city dwellers, allowing for the development of more targeted and effective urban improvement strategies, policies that consider emotional well-being, and public spaces that cater to diverse needs.

C. ABM-GABM: A Combined Approach

An in-depth analysis of the advantages and limitations of ABM and GABM suggests that a hybrid approach integrating both methodologies can offer a more effective and scalable solution.

As proposed by [49], hybrid ABM-GABM can be designed to balance computational efficiency and cognitive complexity. One promising strategy for reducing computational overhead is to delegate low-complexity, rule-based decisions

(such as simple state transitions or predefined interactions) to traditional ABM algorithms, while complex decision-making—involving human-like perception, adaptive reasoning, and contextual interpretation—is handled by GAs, leveraging the capabilities of LLMs.

For example, let's consider a crowd evacuation simulation in a public building, designed with a hybrid ABM-GABM approach. Rule-based ABM logic would be well-suited to govern deterministic actions such as obstacle avoidance, speed controlling, where agents navigate around furniture and walls using predefined pathfinding algorithms without the need for complex reasoning. However, certain human-like decisions necessitate a more adaptive and context-aware approach. For instance, in situations where an agent is trapped in a congested area, it may require a response beyond predefined rules or manually defined objective functions. Instead of following a rigid deterministic reaction, the agent can query the LLM to evaluate possible actions, considering factors like perceived danger, stress levels, and social influence. Accordingly, if an agent comes across an injured individual, it may deliberate on whether to stop and assist or continue toward the exit, with the LLM accounting for social norms, urgency, and the agent's personality traits.

Regarding validation, developers and validators can employ a combination of traditional validation techniques, as shown in Fig. 3, alongside specific methodologies tailored to GABM-based simulations, such as direct human-agent interaction. Traditional methods are particularly effective for assessing aggregated outcomes and macrolevel dynamics, ensuring that the simulation's emergent behaviors align with real-world data and theoretical expectations. Conversely, human-agent interaction methodologies provide a means to evaluate decision-making coherence, consistency, and information diffusion at the microlevel, offering insights into individual agent behavior and localized dynamics.

For example, in a land-uses GABM for an urban environment, data analytics, and visualization can be leveraged to validate population distribution patterns across different city zones, ensuring that simulated crowd densities match empirical data. Meanwhile, agent interviews can be conducted to assess decision-making adaptation based on perception, personality traits, and contextual influences. A concrete example is evaluating how agents dynamically adjust their behavior in response to perceived overcrowding—determining whether individuals seek alternative routes, adapt their schedules, or change their mode of transportation in a way that aligns with real-world human decision-making patterns. This hybrid validation bridges the gap between quantitative validation techniques and qualitative assessments of agent believability.

In conclusion, combining traditional approaches and GAs can effectively mitigate the computational overhead associated with frequent LLM queries, making large-scale simulations more feasible while preserving the advantages of generative adaptability.

X. CONCLUSION

In this work, we explored how the integration of GAs, enhanced by LLMs, expands ABMs by enabling the simulation

of more realistic and adaptive human behaviors. We provided an overview of the emerging GABM field, discussing its potential applications, challenges, and limitations.

A key conclusion of this study is that both traditional ABM and GABM have distinct strengths and weaknesses, making them complementary rather than mutually exclusive. Rather than viewing GABM as a replacement for ABM, we highlighted the need for a balanced approach that integrates GABM's cognitive and adaptive capabilities while preserving the efficiency, scalability, and interpretability of traditional ABM. Our comparative analysis underscores the importance of further research to develop hybrid methodologies that effectively harness the advantages of both paradigms, paving the way for more robust, flexible, and human-like agent-based simulations.

In conclusion, the advent of GAs represents a significant leap forward in ABM, offering opportunities for modeling complex social systems. As the research community continues to refine these technologies and their validation methods, GABMs are poised to become indispensable tools for understanding and addressing real-world challenges, social dynamics, and beyond.

REFERENCES

- [1] D. Dillion, N. Tandon, Y. Gu, and K. Gray, "Can AI language models replace human participants?" *Trends Cogn. Sci.*, vol. 27, no. 7, pp. 2023.
- [2] I. Grossmann, M. Feinberg, D. C. Parker, N. A. Christakis, P. E. Tetlock, and W. A. Cunningham, "AI and the transformation of social science research," *Science*, vol. 380, pp. 1108–1109, 2023, Art. no. 6650.
- [3] J. S. Park, O. Joseph, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," In *Proc. 36th Annu. ACM Symp. User Interface Softw. Technol.*, 2023, pp. 1–22.
- [4] X. Zhou et al., "SOTOPIA: Interactive evaluation for social intelligence in language agents," in *Proc. 12th Int. Conf. Learn. Represent.*, 2024.
- [5] D. Kahneman, "Maps of bounded rationality: Psychology for behavioral economics," *Amer. Econ. Rev.*, vol. 93, no. 5, pp. 1449–1475, 2003.
- [6] J. J. Horton, "Large language models as simulated economic agents: What can we learn from homo silicus?" *Nat. Bur. Econ. Res., Tech. Rep.* 2023.
- [7] J. Zhang, X. Xu, N. Zhang, R. Liu, B. Hooi, and S. Deng, "Exploring collaboration mechanisms for LLM agents: A social psychology view," in *Proc. ICLR 2024 Workshop Large Lang. Model (LLM) Agents*, 2024.
- [8] J. M. Epstein, "Agent-based computational models and generative social science," *Complexity*, vol. 4, no. 5, pp. 41–60, 1999.
- [9] J. M. Epstein, *Generative Social Science: Studies in Agent-Based Computational Modeling*. Princeton, NJ, USA: Princeton Univ. Press, 2012.
- [10] F. L. Shults, R. Gore, W. Wildman, C. Lynch, J. E. Lane, and M. Toft, "A generative model of the mutual escalation of anxiety between religious groups," *J. Artif. Societies Social Simul.*, vol. 21, no. 4, 2018, Art. no. 7.
- [11] B. Gao, W. K. Chan, and X. Deng, "Generative agent-based modeling and empirical validation of the size distribution of hospitals," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 47, no. 11, pp. 3089–3100, Nov. 2016.
- [12] C. Lofdahl, E. Stickgold, B. Skarin, and I. Stewart, "Extending generative models of large scale networks," *Proc. Manuf.*, vol. 3, pp. 3868–3875, 2015.
- [13] T. C. Schelling, "Dynamic models of segregation," *J. Math. Sociol.*, vol. 1, no. 2, pp. 143–186, 1971.
- [14] C. W. Reynolds, "Flocks, herds and schools: A distributed behavioral model," in *Proc. 14th Annu. Conf. Comput. Graph. Interact. Technol.*, 1987, pp. 25–34.
- [15] A. S. Vezhnevets et al., "Generative agent-based modeling with actions grounded in physical, social, or digital space using concordia," 2023, *arXiv:2312.03664*.
- [16] B. Edmonds et al., "Different modelling purposes," *J. Artif. Societies Social Simul.*, vol. 22, no. 3, 2019, Art. no. 6.
- [17] R. Conte and M. Paolucci, "On agent-based modeling and computational social science," *Front. Psychol.*, vol. 5, 2014, Art. no. 83393.

- [18] J. H. Holland, *Adaptation in Natural and Artificial Systems: An Introductory Analysis with Applications to Biology, Control, and Artificial Intelligence*. Cambridge, MA, USA: MIT Press, 1992.
- [19] I. de Zarzà, J. de Curtò, G. Roig, P. Manzoni, and C. T. Calafate, "Emergent cooperation and strategy adaptation in multi-agent systems: An extended coevolutionary theory with LLMS," *Electronics*, vol. 12, no. 12, 2023, Art. no. 2722.
- [20] M. Mitchell, *An Introduction to Genetic Algorithms*. Cambridge, MA, USA: MIT Press, 1998.
- [21] J. H. Holland, "Genetic algorithms," *Sci. Am.*, vol. 267, no. 1, pp. 66–73, 1992.
- [22] Hans, G. Beyer, and H.-P. Schwefel, "Evolution strategies—A comprehensive introduction," *Natural Comput.*, vol. 1, pp. 3–52, 2002.
- [23] E. M. Zechman, "Integrating evolution strategies and genetic algorithms with agent-based modeling for flushing a contaminated water distribution system," *J. Hydroinf.*, vol. 15, no. 3, pp. 798–812, 2013.
- [24] R. G. Reynolds, "An introduction to cultural algorithms," in *Proc. 3rd Annu. Conf. Evol. Program.*, vol. 24. Singapore: World Scientific, 1994, pp. 131–139.
- [25] D. A. Ostrowski, T. Tassier, M. Everson, and R. G. Reynolds, "Using cultural algorithms to evolve strategies in agent-based models," in *Proc. Congr. Evol. Comput. CEC'02 (Cat. No. 02TH8600)*, vol. 1. Piscataway, NJ, USA: IEEE Press, 2002, pp. 741–746.
- [26] C. Cioffi-Revilla, K. De Jong, and J. K. Bassett, "Evolutionary computation and agent-based modeling: Biologically-inspired approaches for understanding complex social systems," *Comput. Math. Org. Theory*, vol. 18, no. 3, pp. 356–373, Sep. 2012.
- [27] Z. Kobti, R. G. Reynolds, and T. Kohler, "A multi-agent simulation using cultural algorithms: The effect of culture on the resilience of social systems," in *Proc. Congr. Evol. Computat. (CEC)*, vol. 3. Piscataway, NJ, USA: IEEE Press, 2003, pp. 1988–1995.
- [28] G. Cornacchia, M. Böhm, G. Mauro, M. Nanni, D. Pedreschi, and L. Pappalardo, "How routing strategies impact urban emissions," in *Proc. 30th Int. Conf. Adv. Geographic Inf. Syst.*, 2022, pp. 1–4.
- [29] L. P. Kaelbling, M. L. Littman, and A. W. Moore, "Reinforcement learning: A survey," *J. Artif. Intell. Res.*, vol. 4, pp. 237–285, 1996.
- [30] K. Arulkumaran, M. P. Deisenroth, M. Brundage, and A. A. Bharath, "Deep reinforcement learning: A brief survey," *IEEE Signal Process. Mag.*, vol. 34, no. 6, pp. 26–38, 2017.
- [31] A. Al-Khayarin and O. Halabi, "Smart 3d simulation of covid-19 for evaluating the social distance measures," in *Proc. HCI Int. 2021-Posters: 23rd HCI Int. Conf., HCII 2021, Virtual Event, Jul. 24–29, Part III 23*. New York: Springer, 2021, pp. 551–557.
- [32] Y. Zhang et al., "Learning decentralized traffic signal controllers with multi-agent graph reinforcement learning," *IEEE Trans. Mobile Comput.*, vol. 23, pp. 7180–7195, Jun. 2024.
- [33] Z. Feng, M. Huang, D. Wu, E. Q. Wu, and C. Yuen, "Multi-agent reinforcement learning with policy clipping and average evaluation for UAV-assisted communication Markov game," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 14281–14293, Dec. 2023.
- [34] F. Li et al., "An agent-based procedure with an embedded agent learning model for residential land growth simulation: The case study of nanjing, china," *Cities*, vol. 88, pp. 155–165, 2019.
- [35] B. Chen, W. Li, and H. Pei, "Deep recurrent q-learning for research on complex economic system," in *Proc. IEEE 5th Inf. Technol. Mechatron. Eng. Conf. (ITOEC)*. Piscataway, NJ, USA: IEEE Press, 2020, pp. 583–588.
- [36] M. Ale Ebrahim Dehkordi, J. M. Lechner, A. Ghorbani, I. Nikolic, E. J. L. Chappin, and P. M. Herder, "Using machine learning for agent specifications in agent-based models and simulations: A critical review and guidelines," *J. Artif. Societies Social Simul.*, vol. 26, no. 1, 2023, Art. no. 9.
- [37] Z. Siyue Ren, R. Cui, Z. Song, and S. Wang, "Emergence of social norms in generative agent societies: Principles and architecture," in *Proc. 33rd Int. Joint Conf. Artif. Intell. (IJCAI)*, 2024.
- [38] P. Schramowski, C. Turan, N. Andersen, C. A. Rothkopf, and K. Kersting, "Large pre-trained language models contain human-like biases of what is right and wrong to do," *Nature Mach. Intell.*, vol. 4, no. 3, pp. 258–268, 2022.
- [39] (A. Fair), Meta Fundamental AI Research Diplomacy Team et al., "Human-level play in the game of diplomacy by combining language models with strategic reasoning," *Science*, vol. 378, no. 6624, pp. 1067–1074, 2022.
- [40] S. Borysov, J. Rich, and F. Pereira, "How to generate micro-agents? A deep generative modeling approach to population synthesis," *Transport. Res. Part C: Emerg. Technol.*, vol. 106, pp. 73–97, 2019.
- [41] N. Ghaffarzadegan, A. Majumdar, R. Williams, and N. Hosseinichimeh, "Generative agent-based modeling: An introduction and tutorial," *System Dyn. Rev.*, vol. 40, no. 1, 2024, Art. no. e1761.
- [42] H. Zhang et al., "Building cooperative embodied agents modularly with large language models," 2023, *arXiv:2307.02485*.
- [43] J. Wang et al., "Global evidence of expressed sentiment alterations during the covid-19 pandemic," *Nature Human Behaviour*, vol. 6, no. 3, pp. 349–358, Mar. 2022.
- [44] D. Torren-Peraire, I. Savin, and J. van den Bergh, "An agent-based model of cultural change for a low-carbon transition," *J. Artif. Societies Social Simul.*, vol. 27, no. 1, 2024, Art. no. 13.
- [45] B. Desmarchelier and E. S. Fang, "National culture and innovation diffusion. exploratory insights from agent-based modeling," *Technol. Forecasting Social Change*, vol. 105, pp. 121–128, 2016.
- [46] C. Gao et al., "Social-network simulation system with large language model-empowered agents," 2023, *arXiv:2307.14984*.
- [47] R. Williams, N. Hosseinichimeh, A. Majumdar, and N. Ghaffarzadegan, "Epidemic Modeling with Generative Agents," 2023, *arXiv:2307.04986*.
- [48] L. R. Goldberg, "An alternative "description of personality": The big-five factor structure," in *Personality and Personality Disorders*. Routledge, 2013, pp. 34–47.
- [49] Z. Kaiya et al., "Lyfe agents: Generative agents for low-cost real-time social interactions," 2023, *arXiv:2310.02172*.
- [50] C. Qian et al., "Chatdev: Communicative agents for software development," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics, in Long Papers*, vol. 1, 2024, pp. 15174–15186.
- [51] L. Wang et al., "A survey on large language model based autonomous agents," *Front. Comput. Sci.*, vol. 18, no. 6, Mar. 2024, Art. no. 186345.
- [52] N. Klapach "The comparative emotional capabilities of five popular large language models," *Crit. Debates Humanities, Sci. Global Justice*, vol. 2, no. 1, 2024.
- [53] E. Y. Chang, "Modeling emotions and ethics with large language models," 2024, *arXiv:2404.13071*.
- [54] Z. Li, G. Chen, R. Shao, D. Jiang, and L. Nie, "Enhancing the emotional generation capability of large language models via emotional chain-of-thought," 2024, *arXiv:2401.06836*.
- [55] T. Liu, J. Yang, and Y. Yin, "Toward llm-agent-based modeling of transportation systems: A conceptual framework," 2024, *arXiv:2412.06681*.
- [56] F. Xu, J. Zhang, C. Gao, J. Feng, and Y. Li, "Urban generative intelligence (UGI): A foundational platform for agents in embodied city environment," 2023, *arXiv:2312.11813*.
- [57] G. Istrate, "Models we can trust: Toward a systematic discipline of (agent-based) model interpretation and validation," in D. Frank, L. Alessio, E. Uille, and N. Ann, Eds., *20th Int. Conf. Autonomous Agents Multiagent Syst., Virtual Event, United Kingdom, May 3–7*. ACM, 2021, pp. 6–11.
- [58] W. Rand and R. T. Rust, "Agent-based modeling in marketing: Guidelines for rigor," *Int. J. Res. Marketing*, vol. 28, no. 3, pp. 181–193, 2011.
- [59] J. Chandler, M. Cumpston, T. Li, M. J. Page, and V. J. H. W. Welch, *Cochrane Handbook for Systematic Reviews of Interventions*. Hoboken: Wiley, 2019.
- [60] I. Lakatos, "History of science and its rational reconstructions," in *PSA: Proc. Biennial Meeting Philosophy Sci. Assoc.*, vol. 1970. Cambridge, U.K.: Cambridge Univ. Press, 1970, pp. 91–136.
- [61] A. Collins, M. Koehler, and C. Lynch, "Methods that support the validation of agent-based models: An overview and discussion," *J. Artif. Societies Social Simul.*, vol. 27, no. 1, 2024, Art. no. 11.
- [62] F. Klügl, "A validation methodology for agent-based simulations," in *Proc. ACM Symp. Appl. Comput. (SAC)*. New York, NY, USA: ACM, 2008, pp. 39–43.
- [63] A. Louloudi and F. Klügl, "Immersive face validation: A new validation technique for agent-based simulation," in *Proc. Federated Conf. Comput. Sci. Inf. Syst. (FedCSIS)*, 2012, pp. 1255–1260.
- [64] W.-L. Chiang et al., "Chatbot arena: An open platform for evaluating LLMS by human preference," 2024, *arXiv:2403.04132*.
- [65] L. Zheng et al., "Judging LLM-as-a-judge with MT-bench and chatbot arena," *Adv. Neur. Inf. Process. Syst.*, vol. 36, pp. 135–141, 2024.
- [66] M. Binz and E. Schulz, "Using cognitive psychology to understand GPT-3," *Proc. Nat. Acad. Sci.*, vol. 120, no. 6, 2023, Art. no. e2218523120.
- [67] R. Shiffrin and M. Mitchell, "Probing the psychology of AI models," *Proc. Nat. Acad. Sci.*, vol. 120, no. 10, 2023, Art. no. e2300963120.
- [68] P. Windrum, G. Fagiolo, and A. Moneta, "Empirical validation of agent-based models: Alternatives and prospects," *J. Artif. Societies Social Simul.*, vol. 10, no. 2, 2007, Art. no. 8.

- [69] G. Serapio-García et al., “Aleksandra Faust, and Maja Matarić,” *Pers. Traits Large Lang. Models*, 2023.
- [70] P. Ekman, “What scientists who study emotion agree about,” *Perspectives Psychol. Sci.*, vol. 11, no. 1, pp. 31–34, 2016. PMID: 26817724.
- [71] A. Bechara, H. Damasio, and A. R. Damasio, “Emotion, decision making and the orbitofrontal cortex,” *Cerebral Cortex*, vol. 10, no. 3, pp. 295–307, 2000.
- [72] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” *Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 9459–9474, 2020.
- [73] L. Huang et al., “A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions,” *ACM Trans. Inf. Syst.*, vol. 43, no. 2, pp. 1–55, 2024.
- [74] S. Mohammadzadeh, J. D. Guerra, M. Bonizzato, R. Rabbany, and G. Farnadi, “Hallucination detox: Sensitive neuron dropout (send) for large language model training,” 2024, *arXiv:2410.15460*.
- [75] L. Ouyang et al., “Training language models to follow instructions with human feedback,” *Adv. Neural Inf. Process. Syst.*, vol. 35, pp. 27730–27744, 2022.
- [76] Y.-S. Chuang, Y. Xie, H. Luo, Y. Kim, J. R. Glass, and P. He, “Dola: Decoding by contrasting layers improves factuality in large language models,” in *Proc. 12th Int. Conf. Learn. Represent.*, 2024.
- [77] R. Cohen, M. Hamri, M. Geva, and A. Globerson, “LM vs LM: Detecting factual errors via cross examination,” in H. Bouamor, J. Pino, and K. Bali, Eds., *Proc. 2023 Conf. Emp. Methods Nat. Lang. Process.*. Singapore: Assoc. Comput. Linguistics, Dec. 2023, pp. 12621–12640.
- [78] R. Liu et al., “Training socially aligned language models on simulated social interactions,” 2023, *arXiv:2305.16960*.
- [79] A. Vaswani, “Attention is all you need,” *Adv. Neur. Inf. Process. Syst.*, vol. 8, pp. 231–241, 2017.
- [80] C. Adornetto, V. Fionda, and G. Greco, On the effectiveness of compact strategies for opinion diffusion in social environments, in *Proc. ECAI*. Amsterdam, The Netherlands: IOS Press, 2023, pp. 11–18.
- [81] Z. Wu et al., “Shall we team up: Exploring spontaneous cooperation of competing LLM agents,” in Y. Al-Onaizan, M. Bansal, and Y.-N. Chen, Eds., *Findings of the Association for Computational Linguistics: EMNLP*. Miami, FL, USA: Assoc. Comput. Linguistics, Nov. 2024, pp. 5163–5186.
- [82] J. Doe “Applications of large language models in urban planning and management,” *J. Urban Technol.*, vol. 7, pp. 140–149, 2023.
- [83] J. M. Epstein, “Inverse generative social science: Backward to the future,” *J. Artif. Societies Social Simul.*, vol. 26, no. 2, 2023, Art. no. 9.
- [84] S. Guo, N. Mokherberian, and K. Lerman, “A data fusion framework for multi-domain morality learning,” in *Proc. Int. AAAI Conf. Web Social Media*, vol. 17, pp. 281–291, 2023.
- [85] J. Xu et al., “Effects of urban living environments on mental health in adults,” *Nat. Med.*, vol. 29, no. 6, pp. 1456–1467, 2023.
- [86] R. Chitrakar, H. Wen, and R. Evans, “Virtual reality simulation of urban environments for human emotion analysis,” *Front. Virtual Reality*, vol. 3, 2022, Art. no. 844501.